



EFAST-2026

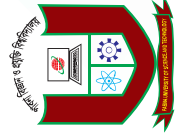
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Faculty of Science, and Life and Earth Science
Pabna University of Science and Technology, Bangladesh



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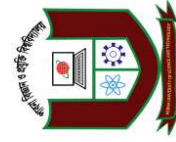
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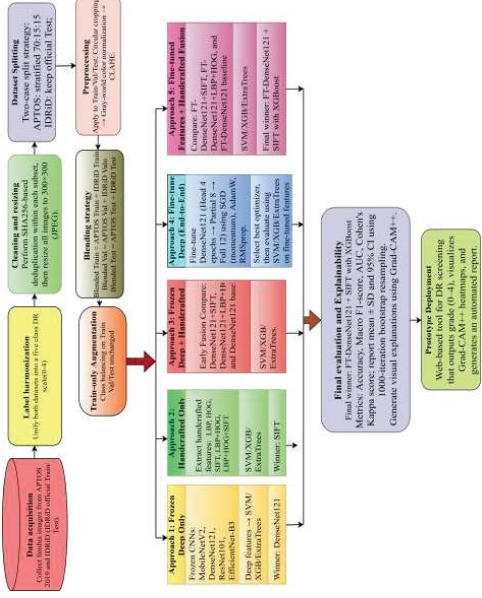
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separability. To support screening-oriented deployment, the selected model was further integrated into an explainable web-based prototype that reports predicted grade, class probabilities, and Grad-CAM++ visualizations. These findings highlight the potential of the proposed framework to support accurate, transparent, and scalable diabetic retinopathy screening in resource-limited settings.



Keywords: Diabetic Retinopathy; Retinal Fundus Images; Deep Learning; Handcrafted Features; Feature Fusion.

Paper ID: 115

Comparative Analysis of Missing Value Imputation Methods with a Hybrid Approach for HCV Prediction Using Machine Learning

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Abstract

Missing values in clinical datasets can substantially affect the reliability, stability, and diagnostic performance of machine learning models, particularly in small and imbalanced medical datasets such as Hepatitis C Virus (HCV) records. This study presents a comparative evaluation of missing-value imputation strategies for HCV prediction and disease-severity modeling using machine learning. The publicly available UCI HCV dataset was used, containing 615 samples and 12 demographic and biochemical predictors, including age, sex, albumin, alkaline phosphatase, alanine aminotransferase, aspartate aminotransferase, bilirubin, cholinesterase, cholesterol, creatinine, gamma-glutamyl transferase, and total protein. The dataset consisted of 540 healthy controls (87.8%) and 75 HCV-related cases (12.2%), representing hepatitis, fibrosis, and cirrhosis stages. Before model development, 31 natural missing values were identified, corresponding to approximately 0.36% missingness, mainly concentrated in ALP and CHOL. Four missing-value handling strategies, namely MICE, Median, Sampling-based imputation, and the proposed Hybrid MICE-Median approach, were evaluated under a leakage-safe analytical workflow using an 80:20 train-test split. Eight machine learning algorithms were assessed for binary HCV classification and disease-severity regression based on ordered HCV stages. Under complete-case analysis, 589 samples remained after excluding incomplete records, and XGBoost achieved the strongest regression performance with $R^2 = 0.827$, MAE = 0.088, and RMSE = 0.270, followed by Random Forest with $R^2 = 0.798$ and RMSE = 0.288. For binary HCV classification, XGBoost also produced the most balanced performance, achieving MCC = 0.914, F1-score = 0.920, Accuracy = 0.984, and ROC-AUC = 0.998. In the



artificial MCAR missingness experiment, the proposed Hybrid approach showed stable reconstruction performance across 5%, 10%, and 20% missingness levels, with MAE values of 9.838, 9.391, and 9.933, respectively, and RMSE values ranging from 24.212 to 27.879. Feature-wise analysis showed that low-variability biomarkers such as CHOL, CHE, ALB, and PROT were imputed more reliably, whereas highly variable clinical markers such as GGT, CREA, ALT, AST, and ALP produced larger reconstruction errors, especially under higher missingness. Overall, the findings suggest that the Hybrid MICE-Median approach provides a robust and clinically plausible balance between adaptive estimation and stability, while ensemble-based models, particularly XGBoost, can improve ML-based HCV screening and disease-severity assessment. External validation using larger and clinically diverse datasets is recommended before practical clinical implementation.

Keywords: Hepatitis C Virus; Missing Value Imputation; Hybrid MICE–Median Imputation; Machine Learning; XGBoost.

Paper ID: 115

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